CARCINOMA BREAST

WHEN TO SUSPECT/ RECOGNISE

Case definition:

Suspect case of Ca Breast (For Level 1, 2, 3 and 4 facility) It is based on clinical grounds. A female patient having a firm or hard lump in the breast with or without axillary lymph nodes, nipple discharge or ulceration over the breast is likely to have Ca breast. Advanced cases may present with symptoms suggestive of bone, brain, liver or lung metastasis. The patient may also have a family history of Ca Breast in the first degree relatives. The group of patients with the following risk factors has a greater chance of developing breast carcinoma.

A. Major Risk Factors:

- Female sex: Ca Breast is very uncommon in the male breast (< 1%).
- Age: Ca Breast is very uncommon before 20y, uncommon before 30y and then the incidence rises. Chances of developing Ca Breast in the 6th decade of life are 1 in 24 as compared to 1 in 257 in the 3rd decade.
- Personal history of contralateral Ca Breast.
- Family history of breast cancer especially in first degree relatives (mother, sister, daughter) more so if they had the cancer when premenopausal or had bilateral cancer.
- Non-invasive cancer (ductal or lobular Ca-in-situ).
- Benign proliferative changes with atypia.

B. Minor Risk Factors:

- Early menarche
- Late menopause
- Nulliparity or first child after age 31 years
- Obesity
- Low dose radiation
- Use of HRT and OC pills
- Alcohol

A detailed **physical examination** should then be performed in the following manner:

Breast examination should be done in a well lit room with respect to patient's privacy and comfort. Patient should be examined in lying, sitting and bending forward position. Look for:

- Asymmetry and skin changes in the two breasts.
- Obvious mass and its characteristics including fixity to skin and underlying structures.
- Nipple discharge, retraction, excoriation or inversion
- Axillary lymph nodes: number, location and fixity
- Lt Supraclavicular lymph node
- Evidence of metastasis: hepatomegaly, bony tenderness, lung metastasis

Probable case of Ca Breast (For Level 3 & 4 facility)

- Suspicion and Clinical Examination should be as above.
- At Level 3 facility an FNAC should be performed from the breast lump to confirm the diagnosis.

  - FNAC may be USG guided.
  - If facility is not available, the same may be requisitioned from a nearby centre where it is available or it may be got done at Level 4.
- For suspicious lumps, USG and mammography is required.
  - If planned, they should be done prior to FNAC.
- In case of negative FNAC but where the clinical suspicion is high, a Trucut or an open biopsy should be resorted to.
- It is made sure that the biopsy incision is such that it can be included in the subsequent surgery.
- Estrogen Receptor (ER)/ Progesterone Receptor (PR) status should also be known at this time from the FNAC/ Trucut biopsy.
- Once the diagnosis is confirmed, the disease is staged.
- Chest X-Ray and USG of the abdomen are done to look for metastatic disease.
  - In case symptomatic with bone pain, an X-ray of the particular bone is done.
  - A bone scan is routinely performed to look for bony metastasis in case of locally advanced / advanced disease.
- The patient is staged as per the UICC TNM classification which is given below (Table 1).
- Depending on the stage of the disease, the patient is offered formal surgery in the form of Modified Radical Mastectomy (MRM) or Wide Local Excision (WLE) followed by Chemotherapy (CT), Radiotherapy (RT) and/or Hormonal Therapy (HT) neo-adjuvant CT followed by surgery or only palliative therapy.
- At Level 3 facility, surgery can be performed and the patient referred to Level 4 for further management. At Level 4 facility complete comprehensive treatment can be given.

INCIDENCE OF THE CONDITION IN OUR COUNTRY

The exact incidence of Breast Cancer in our country is not known. However it is the second commonest malignancy in females after Carcinoma of the Cervix and the commonest malignancy in females in the West. It is common in age group above 40 years.

The overall 5 year survival in developing countries is between 30-50% due to late detection, late reporting, non availability of adequate and appropriate medication and absence of screening programmes.

DIFFERENTIAL DIAGNOSIS

- Benign Breast Disease
- Traumatic fat necrosis of the breast
- Tuberculosis of the Breast

DIAGNOSTIC CRITERIA, INVESTIGATIONS, TREATMENT & REFERRAL CRITERIA

LEVEL 1: AT SOLO PHYSICIAN CLINIC &

LEVEL 2: AT 6-10 BEDDED PRIMARY HEALTH CENTRE

Clinical Diagnosis:

The diagnosis of Breast Cancer cannot be confirmed at this level. However a strong clinical suspicion can be made with the help of a good history and sound clinical examination. Patient having a firm or hard lump in the breast with or without axillary lymph nodes or nipple discharge. The patient may present with symptoms suggestive of bone, brain, liver or lung metastasis. The patient may also have a family history of Ca Breast in the first degree relatives or a history of surgery in the other breast.

Investigations:

The patient should not be investigated but explained the likely possibility and forwarded to the next level.

Treatment:

No treatment needs to be offered at this level. Only dressings need be done if the patient has presented at the stage of a fungating mass lesion.

Referral criteria:

Suspicion of Carcinoma Breast based on history and clinical examination. The patient is forwarded with a case sheet containing history and clinical findings in detail.

LEVEL 3: AT 30-100 BEDDED COMMUNITY HEALTH CENTRE

Clinical Diagnosis:

Same as Level 1 & 2 for a fresh case reporting directly.

Investigations:

1. To Confirm Diagnosis:

FNAC. This is done from the main lump to confirm diagnosis and to ascertain ER / PR status. FNAC may be USG guided. In case FNAC is inconclusive, a 'tru-cut' or 'Open biopsy' may be required to take out more representative tissue for pathological examination. This can be done at a nearby centre where facilities are available or the patient may be referred to Level 4 for the same.

2. To Stage the Disease:

Ultrasound Abdomen. To rule out liver metastasis

X Ray Chest. To rule out lung metastasis

X Rays of particular bone sites depending upon patient symptomatology.

Bone Scan may be done in cases of locally advanced / advanced disease to look for bone metastasis. It is also done if the X-Ray shows a suspicious area or if the alkaline phosphatase is raised.

3. To Work up for Surgery:

Blood. Hb, TLC, DLC, BT, CT, ABORh, urea, creatinine, sugar, LFT with AST, ALT and Alkaline Phosphatase

Urine RE.

ECG.

Treatment:

Early Breast Cancer (EBC) Stage I, II: Patients are offered surgery in the form of Modified Radical Mastectomy (MRM) or Wide Local Excision (WLE) with axillary node sampling. Subsequently based on the pathology report, any patient with lesion more than 1 cm or more than 0.5 cm with unfavourable characteristics like lymphovascular invasion should undergo Chemotherapy (CT) (Chemotherapy Schedule Table 2). A patient with node positivity irrespective of the size of the tumor should undergo CT. Radiotherapy (RT) (RT schedule Table 3) is given if WLE is done or if the margin of resection is positive. Hormonal therapy (HT) in the form of Tamoxifen 20 mg OD for the next 5 years in case of postmenopausal and ER/ PR +ve cases. For CT and RT patient has to be referred to Level 4.

Referral criteria:

Need for Chemotherapy/ Radiotherapy. Most patients in our setting come at a stage with tumors warranting either neoadjuvant or adjuvant chemotherapy. These should be referred to Level 4 either after surgery (Early Breast Cancer) or before (Locally Advanced Breast Cancer). This should always be done with the blocks and slides along with a detailed history and clinical notes.

LEVEL 4: AT 100 OR MORE BEDDED DISTRICT HOSPITAL

Clinical Diagnosis:

Same as Level 3 for a fresh case reporting directly.

Investigations:

Same as Level 3 for a fresh case reporting directly. Confirmation of diagnosis with the blocks and slides in case patient comes with FNAC/ Biopsy/ Surgery.

1. To Confirm Diagnosis:

FNAC. This is done from the main lump to confirm diagnosis and to ascertain ER / PR status. FNAC may be USG guided. In case FNAC is inconclusive, a tru-cut or Open biopsy may be required to take out more representative tissue for pathological examination.

2. To Stage the Disease:

Ultrasound Abdomen. To rule out liver metastasis

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X Rays of particular bone sites depending upon patient symptomatology.

Bone Scan may be done in cases of locally advanced / advanced disease to look for bone metastasis. It is also done if the X-Ray shows a suspicious area.

3. To Work up for Surgery:

Blood. Hb, TLC, DLC, BT, CT, ABORh, urea, creatinine, sugar

Urine RE.

ECG.

Table 1: UICC TNM Classification for Carcinoma Breast (Includes assessment by imaging)

T (Primary Tumor)

- TX Primary tumour cannot be assessed
- T0 No evidence of primary tumour
- Tis Carcinoma in situ or Paget's disease of the nipple with no tumour.
- T1 2 cm or less in greatest dimension
- T2 Tumour more than 2 cm but not more than 5 cm in greatest dimension
- T3 Tumour more than 5cm in greatest dimension
- T4 Tumour of any size with direct extension to chest wall or skin
  - T4a Extension to chest wall
  - T4b Oedema (including peau d orange) or ulceration of the skin of breast or satellite nodules confined to same breast
  - T4c Both T4a and T4b
  - T4d Inflammatory carcinoma

(N) Regional lymph nodes

- NX Regional lymph nodes cannot be assessed (e.g., previously removed)
- N0 No regional lymph node metastasis
- N1 Metastasis to movable ipsilateral axillary lymph node(s)
- N2
  - N2a Metastasis to ipsilateral axillary nodes fixed to one another (matted) or to other structures
  - N2b Metastasis on imaging studies to ipsilateral internal mammary nodes
- N3
  - N3a Metastasis to ipsilateral infraclavicular lymph nodes
  - N3b Metastasis in ipsilateral internal mammary and axillary lymph node(s)
  - N3c Metastasis in ipsilateral Supraclavicular lymph nodes

Distant metastasis (M)

MX Distant metastasis cannot be assessed

M0 No distant metastasis

M1 Distant metastasis

Stage grouping (AJCC Jan 2003)

Stage 0	Tis	N0	M0
Stage I	T1	N0	M0
Stage IIA	T0 T1 T2	N1 N1 N0	M0 M0 M0
Stage IIB	T2 T3	N1 N0	M0 M0
Stage IIIA	T0 T1 T2 T3 T3	N2 N2 N2 N1 N2	M0 M0 M0 M0 M0
Stage IIIB	T4 T4 T4	N0 N1 N2	M0 M0 M0
Stage IIIC	Any T	N3	M0
Stage IV	Any T	Any N	M1

Treatment:

1. Early Breast Cancer (EBC) Stage I, II: Patients are offered surgery in the form of Modified Radical Mastectomy (MRM) or Wide Local Excision (WLE) with axillary node sampling. Subsequently based on the pathology report, any patient with lesion more than 1 cm or more than 0.5 cm with unfavourable characteristics like lymphovascular invasion should undergo Chemotherapy (CT) (Chemotherapy Schedule Table 2). A patient with node positivity irrespective of the size of the tumor should undergo CT. Radiotherapy (RT) (RT schedule Table 3) is given if WLE is done or if the margin of resection is positive. Hormonal therapy (HT) in the form of Tamoxifen 20 mg OD for the next 5 years in case of postmenopausal and ER/ PR +ve cases.

2. Locally Advanced Breast Cancer (LABC) Stage III: Patients are offered Neoadjuvant or anterior CT as per the following protocol. The response is assessed after each cycle. After 4 cycles, if the lesion has shown a favourable response and significant regression, surgery (MRM) is done. Paclitaxel based CT is the given postoperatively. Patients who show 'No Response' or 'Disease Progression' i.e. (increase in size while on CT) are given trial CT regimes. HT is given as for patients with EBC.

3. Advanced Breast Cancer Stage IV: HT is given as for patients with EBC. Toilet (Simple) mastectomy is offered for patients with fungating lesions. Experimental CT protocols can be tried in these patients

Table 2 (Chemotherapy Schedule)

1. CEF (For EBC):

- Inj Cyclophosphamide 500 mg / m^2 IV
- Inj Epirubicin 75 mg / m^2 IV
- Inj 5FU 500 mg / m^2 IV

  Repeat every 28 days for 6 cycles

2. CAF (For LABC):

- Inj Cyclophosphamide 500mg / m^2 IV Day 1
- Inj Adriamycin 50 mg / m^2 IV Day 1
- Inj 5FU 500 mg / m^2 IV Day 1 & 8

  Repeat every 28 days for 4 cycles followed by surgery followed by

- Inj Paclitaxel 175 mg / m^2 IV Day 1
  (2 cycles every 28 days)

Table 3 (Radiotherapy Schedule): 6000 Gy divided between breast, axilla and supraclavicular region. (4500 Gy to Breast if margin positive and after breast conservative energy and 1500 Gy to the supraclavicular region)

SUGGESTED READING

1. Sabiston Text Book of Surgery, edited by CM Townsend Jr, RD Beauchamp, B Mark Evers, KL Mattox; 17th edition. 2005, published by Saunders, Elsevier, Philadelphia, USA. ISBN-13:978-81-8147-950-1.
2. Oxford Text book of Surgery, edited by Sir PJ Morris, WC Wood; 2nd edition.2000, published by Oxford University Press, London. ISBN 0 19 262284 4.
3. Bailey& Love's Short Practices of Surgery, edited by RCG Russel, NS Williams, CJK Bulstrode; 24th edition.2004, published by Arnold Publishers, London. ISBN 0 340 80820 9.
4. ASI Text book of Surgery, edited by A A Hai, R B Shrivastava; 1st edition. 2003, published by Tata McGraw-Hill, New Delhi. ISBN 0-07-462149-1.
5. The New Aird's Companion in Surgical studies, edited by KG Burnand, AE Young, J Lucas; 3rd edition. 2005, published by Elsevier, Philadelphia, USA. ISBN-0443 07 211 6.
6. Cancer. Principles & Practice of Oncology. DeVita Jr Vincent T, Hellman Samuel Rosenberg Steven A. 7th Ed. 2005. Philadelphia. Lippincot Raven ISBN 0 7817 4450 4.